

Betting for Sim-to-Real Performance Certificates

Anonymous Authors

Abstract—Consider a typical *test* of a robot system: one observes a sequence of outcomes concerning some aspect of interest (crash or no crash, tracking error, time to completion), and reports a mean (crash risk, average error, mean time to completion) and, more importantly, an interval guaranteed to contain that mean at a prescribed confidence, referred to as a *performance certificate*. Given expensive real-world trials, the sample size is therefore small, and the certificate is often loose. Now consider the same procedure, except that before each real outcome is revealed, the operator “peeks” at a large bank of simulated results, and places a *bet* on where the real outcome will land. As the real outcomes settle the bets, the operator gains or loses wealth. One’s “trust” over simulators also shifts within the portfolio. This paper develops that idea into a sim-to-real betting certificate framework with three contributions: (i) An algorithm that links a scalable bank of simulators to effective bets, and the accumulated betting wealth to the certificate. (ii) A proof that the returned certificate is *anytime valid*, covering the true mean with the prescribed probability, using any simulator bank. (iii) The guaranteed wealth-regret bounds yield configuration principles for the proposed algorithm and simulator bank design to deliver tight certificates. Experiments across synthetic distributions and real-world robot tests, covering both replayed standardized testing outcomes and online runtime evaluation, show the proposed method narrows the certificate by $51.6\% \pm 16\%$ against classic and state-of-the-art baselines, and by $32.26\% \pm 8\%$ in the extremely limited-sample regime (≤ 30 samples).

Notation: The set of real numbers and positive integers are denoted by $\mathbb{R}$ and $\mathbb{Z}$, respectively. $\mathbb{Z}_k = \{1, \dots, k\}$ for some $k \in \mathbb{Z}$. $\|\cdot\|$ denotes the ℓ_2 -norm. For $A \subset \mathbb{R}$, $\text{hull}(A) = [\inf A, \sup A]$ (i.e., the closed interval hull).

I. INTRODUCTION

Consider accurately, efficiently, and fairly evaluating a certain robot in the *real* world, on *real* hardware, and sometimes in *real* time, concerning a particular aspect of its performance (e.g., risk estimation [1], [2], [3] and empirical reward evaluation of learned policies [4]). A fundamental tension often arises between *what one wants*: a tight statistical bound that quantifies uncertainty about the performance measure at a prescribed confidence level, referred to as a *performance certificate* (e.g., sufficiently tight lower and upper confidence bounds on risk), and *what one can afford*: only a limited number of costly real-world tests from which to construct such a certificate. **The central idea of this paper is to leverage increasingly abundant and inexpensive simulators, through the lens of betting, to facilitate provably guaranteed performance certification while reducing reliance on costly real-world tests.**

Formally speaking, consider a probability distribution P over the measurable space $(\mathcal{X}, \mathcal{F})$, representing the real-world data-generating distribution over the robot and environment conditions encountered during testing (e.g., the object pose in a pick-and-place task [5], [6] and the friction and damping parameters affecting a navigation task [4]). Given a bounded performance scoring function $\psi : \mathcal{X} \rightarrow \mathcal{M} \subset \mathbb{R}$, we define the performance evaluation target as the scalar expectation (e.g., the expected success rate of a manipulator completing a pick-and-place task [7], [6], [1]), along with its variance:

$$\mu^* := \mathbb{E}_{x \sim P}[\psi(x)], \quad \sigma^2 := \mathbb{E}_{x \sim P}[(\psi(x) - \mu^*)^2]. \quad (1)$$

For the remainder of the paper, we take $\mathcal{M} \subseteq [0, 1]$. With proper normalization, this covers performance measures that are naturally bounded (e.g., risk and success rate), as well as measures that may be unbounded in principle but have justifiable limits under a specified testing envelope [3] (e.g., the tracking error of a robot operating within prescribed velocity and workspace limits).

A. Probabilistic Certificate (without “Simulators”)

Given μ^* is unknown and cannot be observed directly, one approximation is through Monte Carlo sampling [8], [9]:

$$\hat{\mu}_t = \frac{1}{t} \sum_{i=1}^t \psi(x_i), \quad x_i \stackrel{\text{i.i.d.}}{\sim} P, \quad (2)$$

whose empirical variance often appears in two forms: (i) the uncorrected $\hat{\sigma}_t^2 := \frac{1}{t} \sum_{i=1}^t [\psi(x_i) - \hat{\mu}_t]^2$, and (ii) the Bessel-corrected, unbiased sample variance $\hat{\sigma}_{t,u}^2 := \frac{1}{t-1} \sum_{i=1}^t [\psi(x_i) - \hat{\mu}_t]^2, t \geq 2$.

Although $\lim_{t \rightarrow \infty} \hat{\mu}_t = \mu^*$ (by the law of large numbers [8], [9]), $\hat{\mu}_t$ may substantively deviate from μ^* within the *finite*, or even worse, extremely limited sampling regime, which is common in robot testing practice [2], [3], [6].

This is where the *certificate* (also referred to as a *bound* interchangeably throughout this paper) fits in as an interval, $\mathcal{C} = [r_l, r_u] \subset \mathbb{R}$, $r_l, r_u \in \mathbb{R}$, $r_l \leq r_u$, providing the range of candidate values for the unknown μ^* . The *width* is defined as $r_u - r_l$ as an indication of the *tightness* of the certificate. In practice, the construction of $\mathcal{C}$ often proceeds in a sampling-based fashion as shown in (2), and hence often takes a *probabilistic* form as

$$\mathbb{P}(\mu^* \in \mathcal{C}) \geq 1 - \alpha. \quad (3)$$

In the remainder of this paper, unless specified otherwise, we take $\alpha = 0.05$ (i.e., a confidence level of 95%).

1) *Concentration inequality based certificate*: Among various methods to construct a probabilistic certificate [10], [11], [12], [9], one way immediately following (2) is to apply a certain *concentration inequality* [10], [9], [11] that guarantees,

The program that allows the exact reproduction of all figures in this paper can be found at the open-source code repository <https://anonymous.4open.science/r/Bet4Sim2Real-Certificate-ED7D>.

Supplementary Material: “Betting for Sim-to-Real Performance Certificate”

IV. PROOFS

A. Proof for Proposition [I](#)

For a fixed candidate $\mu \in \mathcal{M}$ with $\mathcal{F}_{t-1}$ denoting the real observations available before y_t is obtained, the process in Algorithm [I](#) is a non-negative super-martingale under the candidate null hypothesis $H_0(\mu) : \mu^* = \mu$, as $M_0(\mu) = 1$, $\mathbb{E}[M_t(\mu) \mid \mathcal{F}_{t-1}] \leq M_{t-1}(\mu)$ ($t = 2, \dots, T$) and $M_t(\mu) \geq 0$. By applying Ville’s inequality [14] to the process, the acceptance and rejection of the candidates is any-time valid, so we have $\mathbb{P}(\forall t \leq T : \mu^* \in \mathcal{C}_t) \geq 1 - \alpha$.

B. Proof for Lemma [I](#)

At every round, Algorithm [I](#) and the true-moment benchmark observe the same real measure and test the same candidate μ . They thus only differ mean and variance inserted into the stake calculation. The following proof, at a high level, takes three linked steps: (i) moment error changes the stake, (ii) stake error changes the logarithm of the one-round wealth factor, and further (iii) changes the one-round log wealth.

Proof:

Step 1: How moment error changes the stake: Fix a candidate $\mu \in \mathcal{M}$ and a round τ . Before the clip_δ operation, the stake obtained from an inserted mean m and variance v is $\frac{\kappa(m-\mu)}{v+(m-\mu)^2}$. The absolute slope of this expression w.r.t. m (holding the variance fixed) satisfies

$$\left| \frac{\kappa(v - (m - \mu)^2)}{(v + (m - \mu)^2)^2} \right| \leq \frac{\kappa}{v + (m - \mu)^2} \leq \frac{\kappa}{v_{\min}}. \quad (23)$$

Similarly, the absolute slope w.r.t. v is

$$\frac{\kappa|m - \mu|}{(v + (m - \mu)^2)^2} \leq \frac{9\kappa}{16\sqrt{3}v^{3/2}} \leq \frac{9\kappa}{16\sqrt{3}v_{\min}^{3/2}}. \quad (24)$$

The first inequality follows because, as a function of $z = |m - \mu| \geq 0$, the L.H.S. is maximized at $z = \sqrt{v/3}$, equivalently $(m - \mu)^2 = v/3$. This explains the numerical constant in a_v with [\(18a\)](#).

Step 2: Combine the two coordinate changes: Consider the two raw stakes (without clip_δ), first change the inserted mean from m_τ to μ^* while keeping v_τ fixed, and then change the variance from v_τ to σ^2 while keeping μ^* fixed. The total change cannot exceed the sum of those two changes. Moreover, the clip_δ operation cannot enlarge this difference. As a result:

$$|\lambda_\tau(\mu) - \lambda_\tau^*(\mu)| \leq \frac{\kappa}{v_{\min}}|m_\tau - \mu^*| + \frac{9\kappa}{16\sqrt{3}v_{\min}^{3/2}}|v_\tau - \sigma^2|. \quad (25)$$

Step 3: Stake difference changes the one-round log wealth: Since both clipped stakes lie in the same interval, so every λ between them also lie in that interval. Consequently, safe clipping gives $1 + \lambda(y_\tau - \mu) > \delta$. Therefore,

$$\left| \frac{\partial}{\partial \lambda} \log[1 + \lambda(y_\tau - \mu)] \right| = \frac{y_\tau - \mu}{1 + \lambda(y_\tau - \mu)} \leq \frac{y_\tau - \mu}{\delta}. \quad (26)$$

Combining this fact with the stake bound [\(25\)](#) gives

$$\left| \log \frac{1 + \lambda_\tau^*(\mu)(y_\tau - \mu)}{1 + \lambda_\tau(\mu)(y_\tau - \mu)} \right| \leq \frac{y_\tau - \mu}{\delta} \left(\frac{\kappa}{v_{\min}}|m_\tau - \mu^*| + \frac{9\kappa}{16\sqrt{3}v_{\min}^{3/2}}|v_\tau - \sigma^2| \right). \quad (27)$$

Step 4: Sum the one-round bounds to obtain wealth regret: Using the product definitions of $M_t^*(\mu)$ and $M_t(\mu)$, followed by [\(27\)](#), gives

$$\begin{aligned} \mathcal{R}_t^+(\mu) &= \max \left\{ \sum_{\tau=1}^t \log \frac{1 + \lambda_\tau^*(\mu)(y_\tau - \mu)}{1 + \lambda_\tau(\mu)(y_\tau - \mu)}, 0 \right\} \\ &\leq \sum_{\tau=1}^t \left| \log \frac{1 + \lambda_\tau^*(\mu)(y_\tau - \mu)}{1 + \lambda_\tau(\mu)(y_\tau - \mu)} \right| \\ &\leq \sum_{\tau=1}^t (y_\tau - \mu) (a_m|m_\tau - \mu^*| + a_v|v_\tau - \sigma^2|) \end{aligned} \quad (28)$$

The first inequality bounds the positive part of a sum by the sum of the absolute values. The second inequality applies [\(27\)](#) at every round. This proves [\(18a\)](#).

Finally, consider

$$\begin{aligned}
& c_{\text{stab}}^2 [(m_\tau - \mu^*)^2 + (v_\tau - \sigma^2)^2] - [a_m |m_\tau - \mu^*| + a_v |v_\tau - \sigma^2|]^2 \\
&= (a_m^2 + a_v^2) [(m_\tau - \mu^*)^2 + (v_\tau - \sigma^2)^2] - [a_m^2 (m_\tau - \mu^*)^2 + 2a_m a_v |m_\tau - \mu^*| |v_\tau - \sigma^2| + a_v^2 (v_\tau - \sigma^2)^2] \\
&= a_m^2 (v_\tau - \sigma^2)^2 - 2a_m a_v |m_\tau - \mu^*| |v_\tau - \sigma^2| + a_v^2 (m_\tau - \mu^*)^2 \\
&= (a_m |v_\tau - \sigma^2| - a_v |m_\tau - \mu^*|)^2 \geq 0.
\end{aligned}$$

The first equality is due to the definition of $\bar{\mathcal{R}}_t$. Taking square roots proves (I8b).

This completes the proof. ■

C. Proof of Theorem I

Proof:

Step 1: relate the trust-update loss to moment error: Given the observed measure y_τ at round τ , Algorithm I (line 8) adds $\log \mathcal{N}(y_\tau; \mu_k, \sigma_k^2)$ to simulator k 's cumulative score. Thus simulator k 's one-round trust-update loss is $-\log \mathcal{N}(y_\tau; \mu_k, \sigma_k^2)$, and a smaller value leads to more future trust. We have its expectation as

$$\begin{aligned}
\mathbb{E}[(y_\tau - \mu_k)^2] &= \mathbb{E}[(y_\tau - \mu^* + \mu^* - \mu_k)^2] \\
&= \mathbb{E}[(y_\tau - \mu^*)^2] + 2(\mu^* - \mu_k)\mathbb{E}[y_\tau - \mu^*] + (\mu^* - \mu_k)^2 \\
&= \sigma^2 + (\mu^* - \mu_k)^2.
\end{aligned} \tag{29}$$

The first term is σ^2 by definition. The middle term is zeros as $\mathbb{E}[y_\tau] = \mu^*$. We can then compute simulator k 's expected trust-update loss with the loss obtained by the true moment oracle within the same formula as

$$\begin{aligned}
\mathbb{E}[-\log \mathcal{N}(y_\tau; \mu_k, \sigma_k^2) + \log \mathcal{N}(y_\tau; \mu^*, \sigma^2)] &= \frac{1}{2} \log \frac{\sigma_k^2}{\sigma^2} + \frac{\mathbb{E}[(y_\tau - \mu_k)^2]}{2\sigma_k^2} - \frac{\mathbb{E}[(y_\tau - \mu^*)^2]}{2\sigma^2} \\
&= \frac{1}{2} \left[\log \frac{\sigma_k^2}{\sigma^2} + \frac{\sigma^2 + (\mu^* - \mu_k)^2}{\sigma_k^2} - 1 \right].
\end{aligned} \tag{30}$$

Step 2: bound the trust-weighted loss:

Given $y_\tau \in [0, 1]$, $\mu_k \in [0, 1]$, and $v_{\min} \leq \sigma_k^2 \leq v_{\max}$, we have the uniform bound

$$\frac{1}{2} \log(2\pi v_{\min}) \leq -\log \mathcal{N}(y_\tau; \mu_k, \sigma_k^2) \leq \frac{1}{2} \log(2\pi v_{\max}) + \frac{1}{2v_{\min}}. \tag{31}$$

Thus the K trust-update losses lie in an interval of length at most b_{score} at any round. Their negatives, which Algorithm I (line 8) accumulates have the same range.

Substituting Algorithm I's score update into the exponential sum gives

$$\begin{aligned}
\sum_{k=1}^K \exp(\eta L_\tau^k) &= \sum_{k=1}^K \exp(\eta L_{\tau-1}^k) \cdot \exp(\eta \log \mathcal{N}(y_\tau; \mu_k, \sigma_k^2)) \\
&\Rightarrow \frac{\sum_{k=1}^K \exp(\eta L_\tau^k)}{\sum_{j=1}^K \exp(\eta L_{\tau-1}^j)} = \sum_{k=1}^K \frac{\exp(\eta L_{\tau-1}^k)}{\sum_{j=1}^K \exp(\eta L_{\tau-1}^j)} \cdot \exp(\eta \log \mathcal{N}(y_\tau; \mu_k, \sigma_k^2)) = \sum_{k=1}^K \pi_\tau^k \exp(\eta \log \mathcal{N}(y_\tau; \mu_k, \sigma_k^2)) \tag{32} \\
&\Rightarrow \log \frac{\sum_{k=1}^K \exp(\eta L_\tau^k)}{\sum_{j=1}^K \exp(\eta L_{\tau-1}^j)} = \log \sum_{k=1}^K \pi_\tau^k \exp(\eta \log \mathcal{N}(y_\tau; \mu_k, \sigma_k^2)) \leq \eta \sum_{k=1}^K \pi_\tau^k \log \mathcal{N}(y_\tau; \mu_k, \sigma_k^2) + \frac{\eta^2 b_{\text{score}}^2}{8}
\end{aligned}$$

The last inequality is by Hoeffding's lemma, and $\pi_\tau^k > 0$, $\sum_{k=1}^K \pi_\tau^k = 1$.

Sum (32) from $\tau = 1$ through t , recall $L_0^k = 0$ we have

$$\eta \max_{k \in \mathbb{Z}_K} L_t^k - \log K \leq \log \frac{\sum_{k=1}^K \exp(\eta L_t^k)}{K} \leq \eta \sum_{\tau=1}^t \sum_{k=1}^K \pi_\tau^k \log \mathcal{N}(y_\tau; \mu_k, \sigma_k^2) + \frac{t\eta^2 b_{\text{score}}^2}{8}. \tag{33}$$

The term in the middles is the sum of the L.H.S. of (32). We thus have

$$\sum_{\tau=1}^t \sum_{k=1}^K \pi_\tau^k [-\log \mathcal{N}(y_\tau; \mu_k, \sigma_k^2)] - \min_{k \in \mathbb{Z}_K} \sum_{\tau=1}^t [-\log \mathcal{N}(y_\tau; \mu_k, \sigma_k^2)] \leq \frac{\log K}{\eta} + \frac{t\eta b_{\text{score}}^2}{8}. \tag{34}$$

This compares the cumulative trust-weighted loss with the smallest cumulative loss among the bank of K simulators. We now express both sides relative to the loss obtained from the true moments. At round τ , conditioned on earlier observations, (30) gives

$$\mathbb{E} \left\{ \sum_{k=1}^K \pi_{\tau}^k \left[-\log \mathcal{N}(y_{\tau}; \mu_k, \sigma_k^2) + \log \mathcal{N}(y_{\tau}; \mu^*, \sigma^2) \right] \middle| \mathcal{F}_{\tau-1} \right\} = \sum_{k=1}^K \frac{\pi_{\tau}^k}{2} \left[\log \frac{\sigma_k^2}{\sigma^2} + \frac{\sigma^2 + (\mu^* - \mu_k)^2}{\sigma_k^2} - 1 \right]. \quad (35)$$

For every simulator, the minimum over the bank is no larger than that simulator's cumulative excess loss. As a result,

$$\mathbb{E} \left\{ \min_{k \in \mathbb{Z}_K} \sum_{\tau=1}^t \left[-\log \mathcal{N}(y_{\tau}; \mu_k, \sigma_k^2) + \log \mathcal{N}(y_{\tau}; \mu^*, \sigma^2) \right] \right\} \leq t \epsilon_K. \quad (36)$$

Combining the above two we have

$$\frac{1}{t} \sum_{\tau=1}^t \mathbb{E} \left\{ \sum_{k=1}^K \frac{\pi_{\tau}^k}{2} \left[\log \frac{\sigma_k^2}{\sigma^2} + \frac{\sigma^2 + (\mu^* - \mu_k)^2}{\sigma_k^2} - 1 \right] \right\} \leq \epsilon_K + \frac{\log K}{\eta t} + \frac{\eta b_{\text{score}}^2}{8}. \quad (37)$$

Step 3: convert (37) into weighted moment error:

Note Lemma 1 requires the errors in the weighted mean and variance, while (37) bounds the excess trust-update loss. The following inequalities supply the conversion:

$$\sigma^2 \geq \sigma_k^2 \Rightarrow \frac{\sigma^2}{\sigma_k^2} - 1 - \log \frac{\sigma^2}{\sigma_k^2} = \int_1^{\frac{\sigma^2}{\sigma_k^2}} \frac{u-1}{u} du \geq \frac{\sigma^2}{\sigma_k^2} \int_1^{\frac{\sigma^2}{\sigma_k^2}} (u-1) du = \frac{(\sigma^2 - \sigma_k^2)^2}{2\sigma^2 \sigma_k^2} \leq \frac{(\sigma^2 - \sigma_k^2)^2}{2\sigma^4}, \quad (38)$$

and

$$0 < \sigma^2 \leq \sigma_k^2 \Rightarrow \frac{\sigma^2}{\sigma_k^2} - 1 - \log \frac{\sigma^2}{\sigma_k^2} = \int_{\frac{\sigma^2}{\sigma_k^2}}^1 \frac{1-u}{u} du \geq \int_{\frac{\sigma^2}{\sigma_k^2}}^1 (1-u) du = \frac{(\sigma_k^2 - \sigma^2)^2}{2\sigma_k^4} \quad (39)$$

lead to

$$\frac{\sigma^2}{\sigma_k^2} - 1 - \log \frac{\sigma^2}{\sigma_k^2} \geq \frac{(\sigma^2 - \sigma_k^2)^2}{2 \max \{ \sigma_k^2, \sigma^2 \}^2} \geq 0. \quad (40)$$

Substituting the above into (30), using $\sigma_k^2 \leq v_{\max}$, $\sigma^2 \leq v_{\max}$, and $c_{\text{mom}} = 2v_{\max} + 4v_{\max}^2$, one has

$$\begin{aligned} \frac{1}{2} \left[\log \frac{\sigma_k^2}{\sigma^2} + \frac{\sigma^2 + (\mu^* - \mu_k)^2}{\sigma_k^2} - 1 \right] &\geq \frac{(\mu_k - \mu^*)^2}{2v_{\max}} + \frac{(\sigma_k^2 - \sigma^2)^2}{4v_{\max}^2} \\ \Rightarrow (\mu_k - \mu^*)^2 + (\sigma_k^2 - \sigma^2)^2 &\leq \frac{c_{\text{mom}}}{2} \left[\log \frac{\sigma_k^2}{\sigma^2} + \frac{\sigma^2 + (\mu^* - \mu_k)^2}{\sigma_k^2} - 1 \right]. \end{aligned} \quad (41)$$

Because the trust weights are nonnegative and sum to one, the squared error of either weighted moment is no larger than the trust-weighted average of the corresponding individual squared errors:

$$\begin{aligned} (m_{\tau} - \mu^*)^2 + (v_{\tau} - \sigma^2)^2 &\leq \sum_{k=1}^K \pi_{\tau}^k \left[(\mu_k - \mu^*)^2 + (\sigma_k^2 - \sigma^2)^2 \right] \\ \Rightarrow \frac{1}{t} \sum_{\tau=1}^t \mathbb{E} \left[(m_{\tau} - \mu^*)^2 + (v_{\tau} - \sigma^2)^2 \right] &\leq c_{\text{mom}} \left(\epsilon_K + \frac{\log K}{\eta t} + \frac{\eta b_{\text{score}}^2}{8} \right). \end{aligned} \quad (42)$$

Step 4: apply Lemma 1:

$$\overline{\mathcal{R}}_t \leq c_{\text{stab}} \sum_{\tau=1}^t \sqrt{(m_{\tau} - \mu^*)^2 + (v_{\tau} - \sigma^2)^2} \Rightarrow \frac{\mathbb{E}[\overline{\mathcal{R}}_t]}{t} \leq \frac{c_{\text{stab}}}{t} \sum_{\tau=1}^t \mathbb{E} \left[\sqrt{(m_{\tau} - \mu^*)^2 + (v_{\tau} - \sigma^2)^2} \right]. \quad (43)$$

Applying Jensen's inequality separately at each round leads to

$$\begin{aligned} \mathbb{E} \left[\sqrt{(m_{\tau} - \mu^*)^2 + (v_{\tau} - \sigma^2)^2} \right] &\leq \sqrt{\mathbb{E}[(m_{\tau} - \mu^*)^2 + (v_{\tau} - \sigma^2)^2]} \\ \Rightarrow \frac{1}{t} \sum_{\tau=1}^t \mathbb{E} \left[\sqrt{(m_{\tau} - \mu^*)^2 + (v_{\tau} - \sigma^2)^2} \right] &\leq \frac{1}{t} \sum_{\tau=1}^t \sqrt{\mathbb{E}[(m_{\tau} - \mu^*)^2 + (v_{\tau} - \sigma^2)^2]}. \end{aligned} \quad (44)$$

Cauchy-Schwarz applied to the finite sum over rounds gives

$$\frac{1}{t} \sum_{\tau=1}^t \sqrt{\mathbb{E}[(m_\tau - \mu^*)^2 + (v_\tau - \sigma^2)^2]} \leq \sqrt{\frac{1}{t} \sum_{\tau=1}^t \mathbb{E}[(m_\tau - \mu^*)^2 + (v_\tau - \sigma^2)^2]}. \quad (45)$$

Combining (45) and (42) completes the proof. $\blacksquare$

D. Proof for Lemma 2

Proof: By Lemma 2's given definition:

$$r_t^k = \frac{\pi_t^k}{\pi_t^{k'}} = \frac{\exp(\eta L_{t-1}^k)}{\sum_{j=1}^K \exp(\eta L_{t-1}^j)} \frac{\sum_{j=1}^K \exp(\eta L_{t-1}^j)}{\exp(\eta L_{t-1}^{k'})} = \frac{\exp(\eta L_{t-1}^k)}{\exp(\eta L_{t-1}^{k'})} = \exp(\eta(L_{t-1}^k - L_{t-1}^{k'})), \quad (46)$$

$$\begin{aligned} G_t^k &= \frac{r_{t+1}^k}{r_t^k} = \exp\left[\eta((L_t^k - L_t^{k'}) - (L_{t-1}^k - L_{t-1}^{k'}))\right] = \exp\left[\eta((L_t^k - L_{t-1}^k) - (L_t^{k'} - L_{t-1}^{k'}))\right] \\ &= \exp\left[\eta(\log \mathcal{N}(y_t; \mu_k, \sigma_k^2) - \log \mathcal{N}(y_t; \mu_{k'}, \sigma_{k'}^2))\right] \end{aligned} \quad (47)$$

Given r_t^k depends on observation only before round t , G_t^k depends on the fresh observation y_t , and all samples are i.i.d., we have

$$r_{t+1}^k = r_t^k G_t^k \Rightarrow \mathbb{E}[r_{t+1}^k] = \mathbb{E}[r_t^k G_t^k] = \mathbb{E}[r_t^k] \mathbb{E}[G_t^k] \leq \rho \mathbb{E}[r_t^k]. \quad (48)$$

Hence

$$r_1^k = 1 \Rightarrow \mathbb{E}[r_2^k] \leq \rho \Rightarrow \dots \Rightarrow \mathbb{E}[r_t^k] \leq \rho^{t-1}. \quad (49)$$

Moreover, we have

$$\begin{aligned} \sum_{k \neq k', k \in \mathbb{Z}_K} r_t^k &= \sum_{k \neq k', k \in \mathbb{Z}_K} \frac{\pi_t^k}{\pi_t^{k'}} = \frac{1 - \pi_t^{k'}}{\pi_t^{k'}} \Rightarrow 1 - \pi_t^{k'} = \pi_t^{k'} \sum_{k \neq k', k \in \mathbb{Z}_K} r_t^k \leq \sum_{k \neq k', k \in \mathbb{Z}_K} r_t^k \\ &\Rightarrow \mathbb{E}[1 - \pi_t^{k'}] \leq \sum_{k \neq k', k \in \mathbb{Z}_K} \mathbb{E}[r_t^k] \leq (K-1)\rho^{t-1}. \end{aligned} \quad (50)$$

Thus,

$$1 - \pi_t^{k'} \leq 1 \Rightarrow \mathbb{E}[1 - \pi_t^{k'}] \leq 1 \Rightarrow \mathbb{E}[1 - \pi_t^{k'}] \leq \min(1, (K-1)\rho^{t-1}). \quad (51)$$

This completes the proof. $\blacksquare$

E. Proof for Theorem 2

Proof:

Step 1: separate reference error from competitor error: By definition

$$m_\tau - \mu^* = \sum_{k=1}^K \pi_\tau^k \mu_k - \mu^* = \sum_{k=1}^K \pi_\tau^k (\mu_k - \mu_{k'}) + \mu_{k'} \sum_{k=1}^K \pi_\tau^k - \mu^* = \mu_{k'} - \mu^* + \sum_{k \neq k'} \pi_\tau^k (\mu_k - \mu_{k'}), \quad (52)$$

$$v_\tau - \sigma^2 = \sum_{k=1}^K \pi_\tau^k \sigma_k^2 - \sigma^2 = \sum_{k=1}^K \pi_\tau^k (\sigma_k^2 - \sigma_{k'}^2) + \sigma_{k'}^2 \sum_{k=1}^K \pi_\tau^k - \sigma^2 = \sigma_{k'}^2 - \sigma^2 + \sum_{k \neq k'} \pi_\tau^k (\sigma_k^2 - \sigma_{k'}^2). \quad (53)$$

Step 2: bound the competitor contribution: Given (52), (53), and the definitions of h_m , h_v , D_m , and D_v , apply the triangle inequality,

$$|m_\tau - \mu^*| \leq h_m + \sum_{k \neq k'} \pi_\tau^k |\mu_k - \mu_{k'}| \leq h_m + D_m \sum_{k \neq k'} \pi_\tau^k = h_m + D_m(1 - \pi_\tau^{k'}), \quad (54)$$

$$|v_\tau - \sigma^2| \leq h_v + \sum_{k \neq k'} \pi_\tau^k |\sigma_k^2 - \sigma_{k'}^2| \leq h_v + D_v \sum_{k \neq k'} \pi_\tau^k = h_v + D_v(1 - \pi_\tau^{k'}). \quad (55)$$

Apply Lemma 2 and use the definition of S_t , we have

$$\mathbb{E}|m_\tau - \mu^*| \leq h_m + D_m \mathbb{E}[1 - \pi_\tau^{k'}] \leq h_m + D_m \min\{1, (K-1)\rho^{\tau-1}\} \Rightarrow \frac{1}{t} \sum_{\tau=1}^t \mathbb{E}|m_\tau - \mu^*| \leq h_m + \frac{D_m S_t}{t}. \quad (56)$$

$$\mathbb{E}|v_\tau - \sigma^2| \leq h_v + D_v \mathbb{E}[1 - \pi_\tau^{k'}] \leq h_v + D_v \min\{1, (K-1)\rho^{\tau-1}\} \Rightarrow \frac{1}{t} \sum_{\tau=1}^t \mathbb{E}|v_\tau - \sigma^2| \leq h_v + \frac{D_v S_t}{t} \quad (57)$$

Step 3: apply Lemma I:

$$\begin{aligned} \overline{\mathcal{R}}_t &= \sup_{\mu \in \mathcal{M}} \mathcal{R}_t^+(\mu) \leq \sum_{\tau=1}^t (a_m |m_\tau - \mu^*| + a_v |v_\tau - \sigma^2|) \\ &\Rightarrow \frac{\mathbb{E}[\overline{\mathcal{R}}_t]}{t} \leq \frac{a_m}{t} \sum_{\tau=1}^t \mathbb{E}|m_\tau - \mu^*| + \frac{a_v}{t} \sum_{\tau=1}^t \mathbb{E}|v_\tau - \sigma^2| \\ &\stackrel{(56), (57)}{\Rightarrow} \frac{\mathbb{E}[\overline{\mathcal{R}}_t]}{t} \leq a_m \left(h_m + \frac{D_m S_t}{t} \right) + a_v \left(h_v + \frac{D_v S_t}{t} \right). \end{aligned} \quad (58)$$

This proves (22).

Step 4: Bound S_t : By the definition of S_t (let $\lceil \cdot \rceil$ denote the ceiling function returning the smallest integer greater than or equal to the enclosed value):

$$\begin{aligned} S_t &= \sum_{\tau=1}^t \min\{1, (K-1)\rho^{\tau-1}\} \leq \sum_{\tau=1}^{1+\lceil \frac{\log(K-1)}{-\log \rho} \rceil} 1 + \sum_{\tau=2+\lceil \frac{\log(K-1)}{-\log \rho} \rceil}^{\infty} (K-1)\rho^{\tau-1} \\ &= 1 + \left\lceil \frac{\log(K-1)}{-\log \rho} \right\rceil + \frac{(K-1)\rho^{1+\lceil \frac{\log(K-1)}{-\log \rho} \rceil}}{1-\rho} \\ \text{Note: } \left\lceil \frac{\log(K-1)}{-\log \rho} \right\rceil &\geq \frac{\log(K-1)}{-\log \rho} \Rightarrow (K-1)\rho^{1+\lceil \frac{\log(K-1)}{-\log \rho} \rceil} \leq \rho \\ &\Rightarrow S_t \leq 1 + \left\lceil \frac{\log(K-1)}{-\log \rho} \right\rceil + \frac{\rho}{1-\rho} \leq 1 + \left\lceil \frac{\log(K-1)}{-\log \rho} \right\rceil + \frac{1}{1-\rho}, \end{aligned} \quad (59)$$

which proves (21). This completes the proof. ■

V. EXPERIMENT DETAILS

This section provides more details (particularly on **C2** and **C3**) complementing the main paper's empirical results in Section III as well as the released code base supporting the reproducibility of the presented results.

A. C2 - command tracking

The real-world samples are collected using a customized software bridge that translates joystick actions from a Logitech F310 gamepad into GR00T compatible commands x^c (linear and yaw) at 50 Hz. For each sample, the velocity error is computed as the difference between the command velocity sent through the gamepad and the actual velocity measured by the OptiTrack Motion Capture System (OTS). The lower body controller runs at a fixed control frequency (50 Hz), and the velocity measurements are sampled at the same frequency.

To establish the paired correspondence between the real-world and simulation data required by SureSim, we replay the same commands recorded from the real-world samples in MuJoCo environment. The resulting paired velocity error is computed from the difference between the replayed command and the measured robot state x^r . For real-world samples, x^r is obtained from the Motion Capture System, while for simulated samples, x^r is obtained from the corresponding robot state in MuJoCo. For the `Sim_MuJoCo_3`, we construct a diverse simulation bank by varying the physical parameters of the MuJoCo environment.

In addition, the implementation of SureSim is adapted from the opensource code [\[1\]](#). Details of the adaptation can be found in the released code base.

B. C3 - peg-in-hole

Continuous mobile manipulator benefits for large, curved, and complex workpieces in many industrial and unstructured environments. However, such flexibility also introduces additional source of performance uncertainty, preventing stringent pose repeatability and accuracy. To identify and quantify this uncertainty, the Configurable Mobile Manipulator Apparatus (CMMA) was developed in [5] to provide a standardized set of target locations for evaluating the performance of the mobile manipulator during continuous motion. As illustrated in Fig. 7 the Type B CMMA used in this study consists of two sides, each containing six retro-reflective fiducials of different sizes. The mobile manipulator system consists of an autonomous

¹<https://github.com/irom-princeton/rapid-policy-evaluation>

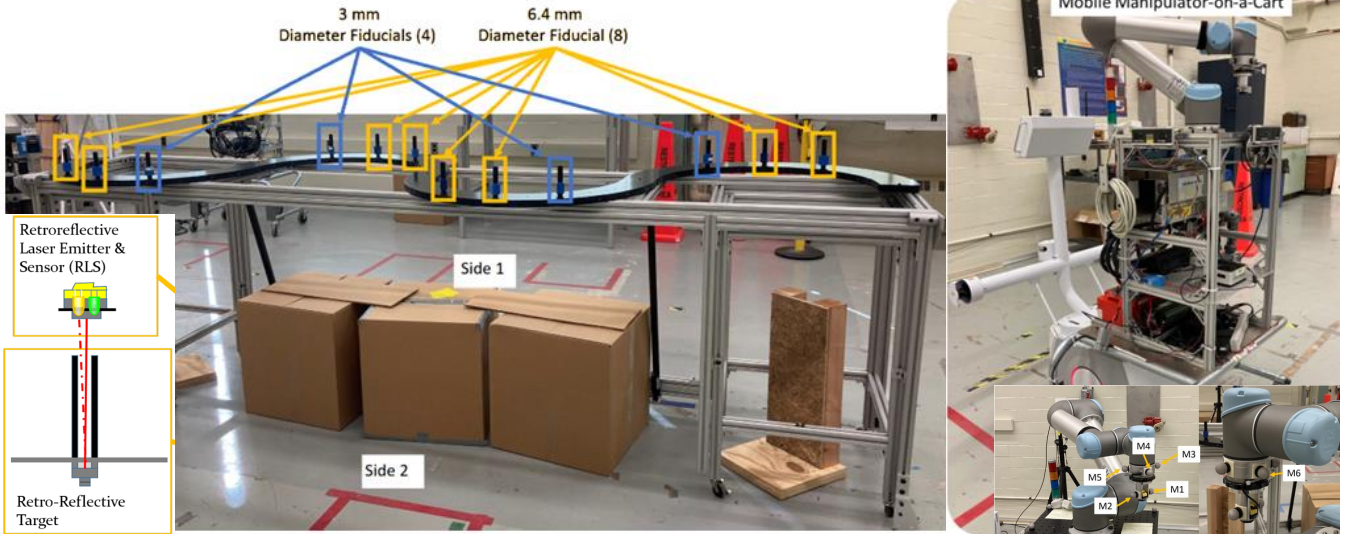


Fig. 7: Overview of the experimental setup. The left side shows the Type B CMMA, while the right side shows the mobile manipulator mounted on the AMR-CT. The left corner highlights the Retro-reflective Laser Sensor and Emitter (RLS), and the right corner shows the optical tracking system (OTS) markers mounted on the end effector of the manipulator. *Note: the figure is adapted from [5]*

mobile robot cart transporter (AMR-CT) and a manipulator, with a Retro-reflective Laser Sensor and Emitter (RLS) mounted at the end effector of the manipulator to detect and intercept these fiducials as the mobile manipulator moves along the apparatus. The AMR-CT uses manufacturer-provided mapping and navigation software together with custom ROS-based velocity control to drive the platform along the CMMA.

In this experiment, the fiducial-interception task serves as a standardized surrogate for a peg-in-hole-type manipulation task that the end-effector must continuously locate and align with a designed target and successfully intercept it while both the AMR-CT and the manipulator are in motion. The experiments design consisted of a full 2^3 factorial experiment with three factors: (i) the search algorithm (varied between Deterministic Spiral Search and a stochastic Unscented Kalman Filter (UKF)-based search), (ii) the mobile-base translation speed (either 0.01 m/s or 0.025 m/s), and (iii) the CMMA side (either Side 1 or Side 2). The order of the eight factor combinations was randomized within each replicate to reduce potential ordering effects. The original NIST study analyzed the experimental data primarily using a 2^3 factorial Analysis of Variance, using p-values to assess the statistical significance of the metrics, along with other prior analysis. Two versions of the NIST dataset are publicly available, released in February² and June³, respectively. In this work, we use the June dataset which has six replicate trials for each of the eight experimental conditions, resulting in a total of $6 \times 8 = 48$ experimental runs.

The performance in this report is evaluated through two primary measurements: (i) the fiducial interception rate, defined as the percentage of target fiducials successfully intercepted during one run and (ii) the average fiducial search time, defined as the mean time required to locate and intercept an individual target, capped at the maximum time allotted to search for that target before the vehicle carries it out of range. In addition, we derived an additional continuous measurement from the recorded data: the distance between the fiducial target and end manipulator end-effector measured by OTS. Details of the computation and data processing procedure are provided in the code base.

C. C3 - pushover

The pushover stability test of the Unitree Go2 robot follows a standard testing method currently being developed as part of WK86916 by the subcommittee ASTM F45.06 Legged Robot Systems.

The impactor is a 50 mm-diameter, 500 g steel ball complies with GB4943, GB4906, IEC60335, IEC60950, IEC60598, UL507, concerning the impact resistance performance of various electronic appliances. It swung as a pendulum from a constant vertical height to impact the enclosure of the subject robot. The subject is a Unitree Go2 EDU quadruped robot with commercial software stack capable of in-place standing (undisturbed) (firmware version anonymized). Across all impact tests, the subject robot starts from the same stationary foot placement (25×40 cm) in contact with the ground. However, its body pose still varies due to the intrinsic design of the robot controller. The impact area therefore still differs from trial to trial, which contributes in part to the uncertainty of the performance measure.

²<https://data.nist.gov/od/id/mds2-3061>

³<https://data.nist.gov/od/id/mds2-3187>

VI.

NOMENCLATURE

$\mathcal{F}$	Collection of measurable events on $\mathcal{X}$
$\mathcal{M}$	The range of performance measurement
$\mathcal{X}$	Sample space
$\psi(\cdot)$	Performance evaluation function
P	Real-world probability distribution
x_t	One state sample(the t -th sample)
$(M_t(\mu))_{t \geq 0}$	The e-process, defined in (7) (i.e., the supermartingale and the wealth in betting of candidate μ)
$1 - \alpha$	The confidence level
α	The significance level
$\delta(\cdot, \cdot)$	The concentration inequality specific function
γ	The half width of the certificate
$\hat{\mu}_t$	The Monte Carlo estimator from t samples in (2)
$\hat{\sigma}_t^2$	The empirical variance
$\hat{\sigma}_{t,u}^2$	The Bessel-corrected, unbiased sample variance
κ	The kelly fraction, controlling the aggressiveness of betting
$\lambda_t(\mu)$	A predictable factor and the stake played in betting of candidate μ at round t calculated through (9)
$\mathcal{C}$	Certificate and bound
$\mathcal{F}_{t-1}$	Filtration, information available up to round $t - 1$
μ	The candidate value composing certificate $\mathcal{C}$
μ^*	The true mean performance in (1)
σ^2	The true variance of performance
$\mathcal{T}_t(\mu)$	The statistic of t -test
H_0	The null hypothesis
r_u, r_l	The upper bound and the lower bound of the certificate
$t_{1-\frac{\alpha}{2}, t-1}$	Quantile of Student's t distribution specified by its significance level and degrees of freedom
$z_{1-\alpha/2}$	Quantile of the standard normal distribution
ϵ_K	The expected Gaussian-loss difference between the best fixed simulator (in the bank) and the true moment pair
η	The learning rate
$\mathbb{E} [\overline{\mathcal{R}}_t] / t$	The expected average worst-candidate positive regret as in (17)
$\mathcal{R}_t(\mu)$	The wealth regret in (15)
$\mathcal{R}_t^+(\mu)$	The positive part of the wealth regret
μ_k, σ_k^2	The mean and variance of the k -th simulator in the bank
$\overline{\mathcal{R}}_t$	The largest wealth regret among candidates μ
π_t^k	The weight assigned to the k -th simulator at round t
ρ	The constant used in 2
$\{Q_\theta\}$	The bank of parameterized simulators
$a_m, a_v, c_{\text{stab}}$	Constants representing the sensitivity of wealth regret to mean error and variance error
b_{score}	A constant reflecting the fluctuation of simulators
c_{mom}	A constant used in (1)
D_m, D_v	The maximum difference of mean and variance between the reference simulator and other simulator in the bank
G_t^k	The trust ratio change after observation t
h_m, h_v	The absolute mean and variance difference between the reference simulator and the true performance
K	The number of simulators in the simulator bank
k	The simulator index in the bank
k'	One reference simulator in the simulator bank
L_t^k	The Gaussian score of the k -th simulator at round t
m_τ, v_τ	The running moments of the simulator bank at step τ
m_t, v_t	The running moments of the simulator bank at step t
$M_t^*(\mu)$	The "ideal" wealth with known true moments
Q	The simulator induced distribution
r_t^k	The trust ratio of reference simulator and any other simulator in the bank
S_t	An intermediate value in (21), indicating the rate of simulator selection or convergence
T	The number of samples in real world
t	The number of observations used in betting process

$v_{\max}$	The maximum variance among all the simulators in the bank
$v_{\min}$	The minimum variance among all the simulators in the bank
y_t	The performance measure of sample x_t
x^c	The command sent to robot in C2 experiment
x^r	The measured velocity of robot in C2 experiment